

AMENDMENT NO. 1 APRIL 1990
TO
IS 4522 : 1986 SPECIFICATION FOR HEAT
RESISTANT ALLOY STEEL AND NICKEL
BASE CASTINGS

(Second Revision)

(Pages 8 and 9, clauses 13, 13.1, 13.2 and 13.3) — Substitute the following for the existing clauses:

'13 REPAIR OF CASTINGS

13.1 Unless otherwise specified by the purchaser in the enquiry and order, castings may be rectified by welding. All repairs by welding shall be carried out in accordance with the procedure laid down in IS 5530 : 1986. If castings have been subjected to non-destructive testing by agreement between the purchaser and the manufacturer, the castings shall be re-examined in the area of repair following any rectifying operation performed on the castings.

13.2 To form the basis of an agreement between the purchaser and the supplier in this respect, the following classification shall apply concerning the extent of repair:

- a) Weld repair involving a depth not exceeding 20 percent of wall thickness or 25 mm, whichever is lower, shall be termed as a minor repair.
- b) Any weld repair exceeding the above shall be termed as a major repair. Further any single repair having an area exceeding 250 mm² for every millimetre of wall thickness shall also be deemed to be a major repair, regardless of the considerations mentioned in (a) above.

13.3 Carbon Equivalent — Unless otherwise specified in the enquiry and order, or otherwise agreed to, the Carbon Equivalent (C. E.) for the purpose of guidance in determination of the pre- and post-weld treatment applicable to carbon and low alloy steels shall be computed as follows:

NOTE — In case of high alloy steels the formula for computing Carbon Equivalent shall be subject to agreement between the parties.

$$\text{C. E.} = \text{C} + \frac{\text{Mn}}{6} + \frac{\text{Cr} + \text{Mo} + \text{V}}{5} + \frac{\text{Ni} + \text{Cu}}{15}$$

13.4 The welding procedure to be followed for any welding that may be required on the surface hardened area shall be as agreed mutually.'

(MTD 17)